Scattering in Different Frames

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0.1 Background

The collision experiment is used to as a tool for understanding and investigating the the structure of atomic and subatomic particles. Analyzing the distribution of scattered projectiles after a collision we can learn information about the target in question. The most widely known experiment was done by Ernest Rutherford (1871-1937) in which he fired streams of α particles at a thin layer of gold atoms. Measuring the distribution of the scattered α particles they deduced that most of the mass of an atom comes from the nucleus.

These experiments are not confined to structures this small, this same idea applies to moving objects like pool balls on a table. Although this is an option most of the these problems correspond to structures we can not see with our eyes. Usually these types of problems are done using quantum mechanics for collision theory, however, many of the key concepts in understanding these problems involve lab and center of mass frames as well as differential cross sections, making them suitable for a classical mechanics approach. These types of problems require a careful attention to detail when analyzing the behavior between frames, that is being able to go back and forth between a lab frame and a center of mass frame (CM)

In order to understand these types of problems, a few concepts must be understood. In example figure 1 we have a simple two particle system in which one particle collides with a stationary one. Here $\mathbf{P}$ denotes the momenta of a particle before the collision and $\mathbf{P}'$ is the momenta after the collision. θ_{cm} is the angle that corresponds to the initial projectiles degree of movement from the collision site in the CM frame and θ_{lab} is the degree of movement in the lab frame. Notice due to conservation of momentum, all $\mathbf{P}$ have the same magnitude in the CM frame.

One other value is important to understand before attempting these types of problems. Thus introducing the **mass ratio** in which equation 1 denotes the ratio of the masses. This value will prove to be important in the steps to come as we've seen earlier in this class. Using these basic principles we can break down the problem proposed using figure 2 where we have a situation were an initially stationary target is struck by a larger projectile being viewed in the lab frame.

$$\lambda = \frac{m_1}{m_2} \tag{1}$$

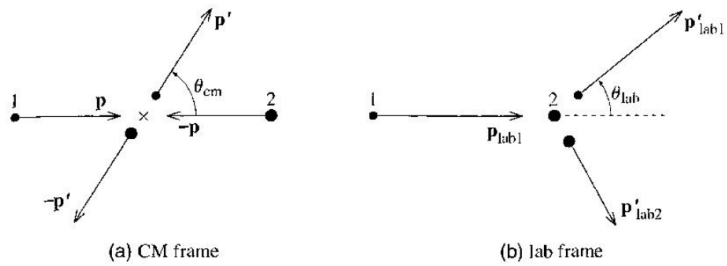

Figure 1: (a) An elastic collision as seen in the CM frame. The two particles approach the CM with equal and opposite momenta where particle 1 is larger than particle 2. (b) The same collision as seen in the lab frame, where particle 2 is initially at rest[1].

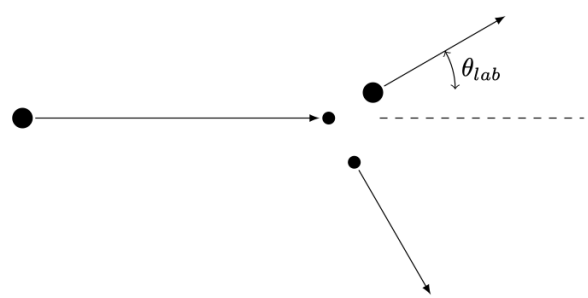

Figure 2: In this situation we have an initially stationary target struck by a more massive projectile as seen from the lab frame.[1]

0.2 Problem

First in order to understand that momenta after the collision, we need to be able to see how the angle θ_{cm} and θ_{lab} relate. To do this, see figure we can make an analogue diagram of this collision comparing the lab frame and CM frame in one, where the lines BC and BD are the initial and final momenta of particle 1 in the CM frame and lines AC and AD are their lab frames. Using the mass ratio we can see that $\mathbf{P}'_{labl}$

$$\mathbf{P}'_{labl} = \lambda \mathbf{P} + \mathbf{P}' \tag{2}$$

With this information we can find the two center of mass angles that correspond to when θ_{lab} is zero. To do this we will need to use the relationship

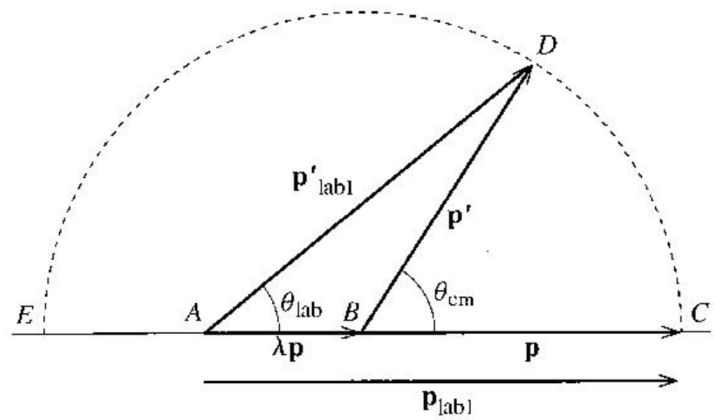

Figure 3: This shows the relationship of initial and final momenta in the CM and lab frame. λ is the mass ratio [1]

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from the right triangle in figure 3 giving us the equation 3. If we set θ_{lab} to zero we can see that θ_{cm1} is either going to be 0 or π. In the center of mass frame, both particle 1 and 2 have the same magnitude of momentum thus θ_{cm2} will have the same values.

$$tan\theta_{lab1} = \frac{sin(\theta_{cm1})}{\lambda + cos(\theta_{cm1})} \tag{3}$$

$$tan(0) = \frac{sin(\theta_{cm1})}{\lambda + cos(\theta_{cm1})} \tag{4}$$

This result might seem ambiguous but by looking closer at figure 3 we can see that in all cases besides θ_{cm} equals 0 or π, θ_{lab} will always be smaller. If θ_{cm} is 0 the two masses coincide with the same path. When θ_{cm} is equal to π, θ_{lab} shares this same angle which results a 0.

We can find an expression for θ_{lab} as a function of θ_{cm} and λ using this same equation giving us that θ_{lab} is simply,

$$\theta_{lab} = tan^{-1}(\frac{sin(\theta_{cm})}{\lambda + cos(\theta_{cm})}) \tag{5}$$

where we see the relationship and can plot θ_{lab} as a function of θ_{cm} for different cases when the mass ratio has different values. Figure 4 illustrates

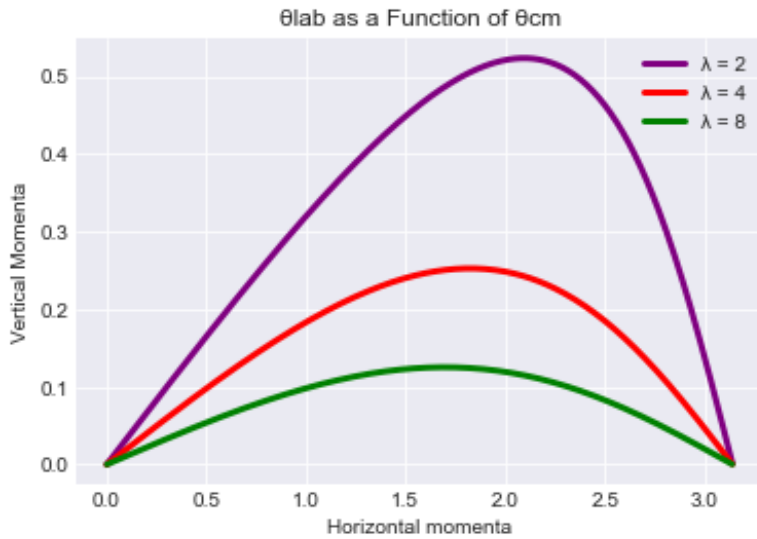

Figure 4: θ_{lab} as a function of θ_{cm} for when λ = 2,4 and 8

this as each color represents a larger mass ratio. This problem is specifically focusing on when λ equals 2. While this graph looks interesting, we really want to find out the maximum possible value of θ_{lab} for a given value of λ. to find this value we will use the same strategy as seen in previous chapters in which we differentiate to solve for our maximum values. Differentiating θ_{lab} with respect to θ_{cm} we get the expression,

$$\frac{\lambda cos(\theta_{cm}) + 1}{\lambda^2 + 2\lambda cos(\theta_{cm}) + 1} = 0 \tag{6}$$

solving for θ_{cm} we get several possible values corresponding to the root of some λ relationship. Not all of these values work for our situation so by testing these values we can match what expression best represents the peak in figure 4. Doing this we get equation 7 and we can plug in values for λ to get a maximum.

$$2tan^{-1}\sqrt{\frac{\lambda + 1}{\lambda - 1}} \tag{7}$$

In order to obtain the lab differential cross-section , we first need to adjust our expression for θ_{lab}. Using the relationship

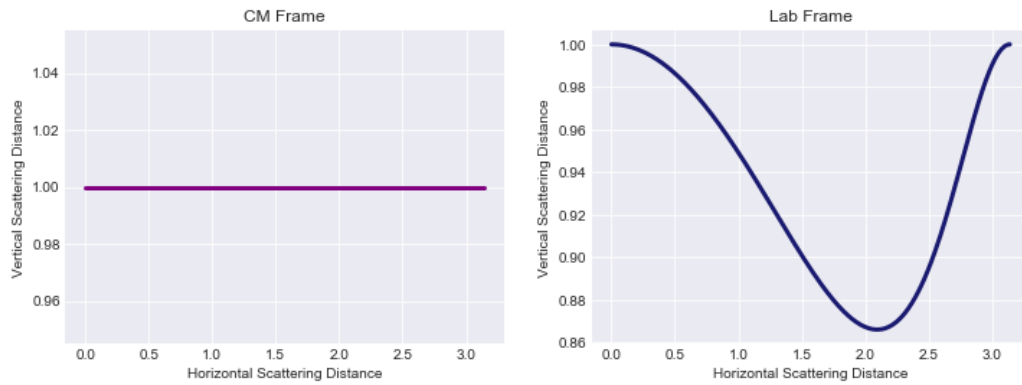

Figure 5: The differential cross section for scattering off a hard sphere is isotropic in the CM frame. In the lab frame it peaks negatively due to the projectile mass being greater than the resting mass.

$$1 + tan^2(\theta) = sec^2(\theta) \tag{8}$$

we can rewrite θ_{lab} in equation 3 to be

$$cos(\theta_{lab}) = \sqrt{\lambda^2 + 2\lambda cos(\theta_{cm}) + cos^2(\theta_{cm})} \tag{9}$$

which we'll use to calculate our cross-section for θ_{lab}.

The final equation we need comes from the relationship between θ_{lab} and θ_{cm} in which we can show the cross-section as,

$$\frac{d(cos\theta_{lab})}{d(cos\theta_{cm})} \tag{10}$$

thus giving the expression for the cross-section 11 and we can plot this result as a function of θ_{lab} as in figure 5

$$\frac{1}{\sqrt{1 + \frac{sin^2(\theta_{cm})}{(\lambda + cos(\theta_{cm}))^2}}} \tag{11}$$

Bibliography

[1] "Classical mechanics john r. taylor : Free download, borrow, and streaming."